

Global Comparison Report

Monte Carlo Strategy Paired-Difference Analytics

GENERATED	MONTE CARLO RUNS	STRATEGIES TESTED
10 May 2026 16:51	10	2
METHOD	UNIT	REPORT TYPE
Discrete-Event Simulation (DES)	Haemodialysis Centre	Comparative Performance Analysis

Abstract

This report presents the results of a Monte Carlo discrete-event simulation study evaluating scheduling strategies for a haemodialysis unit. Stochastic variability in patient volume, machine availability, session duration, and nursing resources is modelled across 10 independent runs per strategy. Distributional summaries, tail-risk metrics (p90 / p99), and head-to-head strategy comparisons are provided to support operational decision-making.

Keywords: Monte Carlo | Discrete-Event Simulation | Haemodialysis | Scheduling | Tail-Risk

1. Simulation Setup

The parameters below fully specify the stochastic environment sampled during each Monte Carlo iteration. All stochastic quantities are drawn independently per run using a seeded Mersenne Twister PRNG.

TEMPORAL BOUNDS

Parameter	Value	Unit	Description
Shift Duration	360	min	Hard wall -- 6 h total shift
Nominal Session Duration	210 - 240	min	Prescribed per-patient session (~4 h target)
Min Viable Session	180	min	Sessions shorter than this are marked failed (3 h)
Machine Cooldown	60	min	Mandatory inter-session machine rest period

RESOURCE RANGES (uniform draw per iteration)

Parameter	Value	Unit	Description
Total Machines	15 - 20	units	Physical dialysis stations available
Patient Volume	15 - 20	patients	Demand sampled each shift
Nursing Staff	2 - 4	nurses	Concurrent supervision capacity
Session Duration	210 - 240	min	Per-patient dialysis time
Machine Ready Delay	0 - 90	min	Post-cooldown preparation lag

STOCHASTIC EVENT GENERATORS

Parameter	Value	Unit	Description
Patient Arrival Offset	Uniform[0, 60]	min	Arrival jitter relative to shift start
Setup Duration	Uniform[10, 20]	min	Machine preparation time per patient

PROBABILITY METRICS

Parameter	Value	Unit	Description
Machine Defect Probability	15.0%	--	Independent Bernoulli draw per machine per session

EXPERIMENT PARAMETERS

Parameter	Value	Unit	Description
Monte Carlo Iterations	10	runs	Independent replications per strategy

2. Global Statistical Summary

Per-strategy distributional metrics (mean +/- SD, p50, p90, p99) aggregated across the full Monte Carlo suite. The p90 and p99 columns serve as tail-risk indicators; lower is better.

FIFO

Metric	Mean +/- SD	p50	p90	p99
Mean Wait Time (min)	28.7 +/- 12.8	27.7	47.3	51.3

Max Wait Time (min)	50.0 +/- 19.8	47.5	80.2	81.8
Avg Session Time (min)	225.3 +/- 3.2	225.2	228.2	231.0

FIXED

Metric	Mean +/- SD	p50	p90	p99
Mean Wait Time (min)	55.3 +/- 20.1	50.3	79.6	87.5
Max Wait Time (min)	107.6 +/- 25.8	98.0	146.2	147.8
Avg Session Time (min)	222.4 +/- 5.6	223.9	228.3	228.6

3. Comparative Analytics

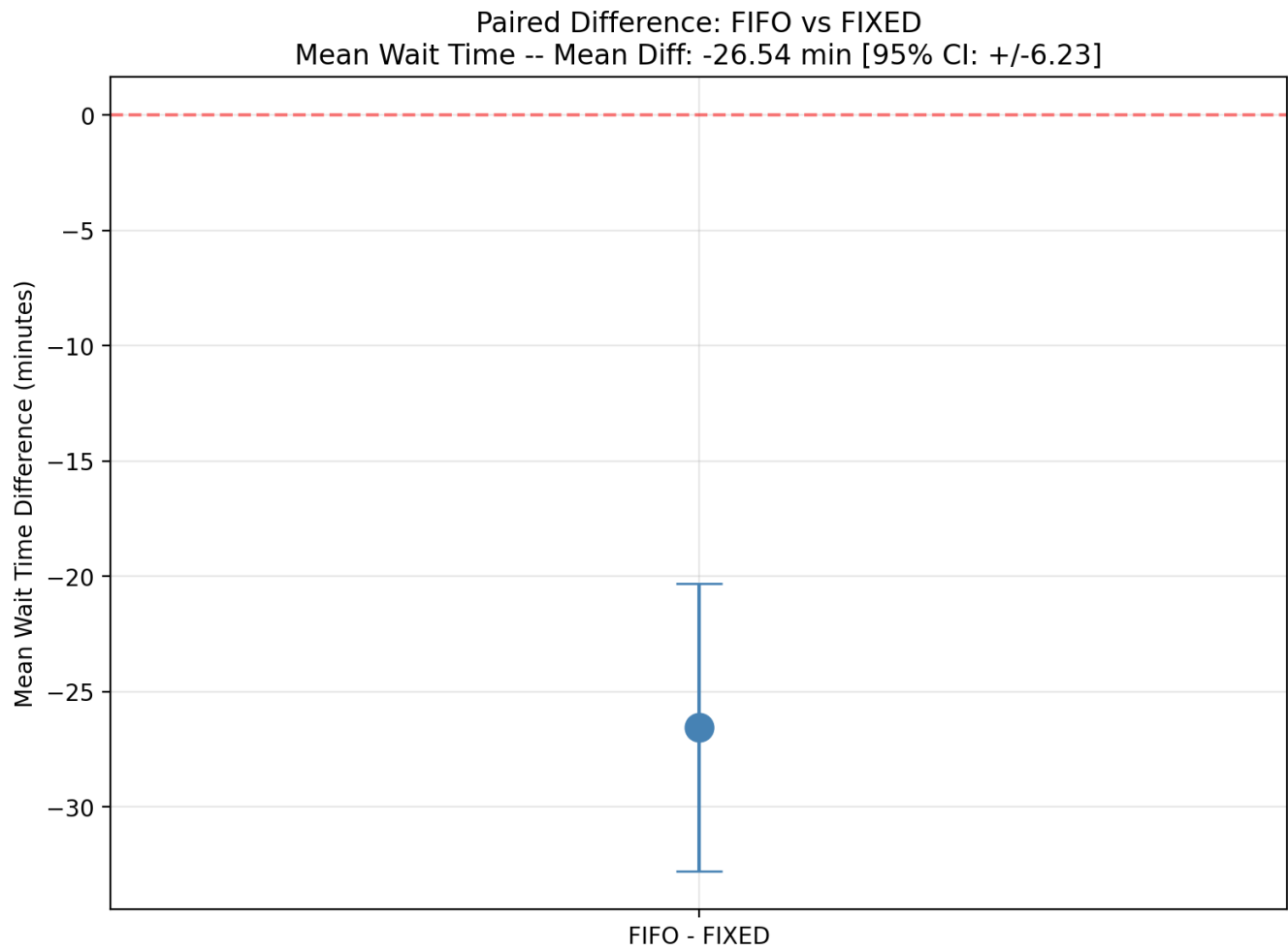


Figure 1. Paired-difference analysis of mean wait time across strategies.

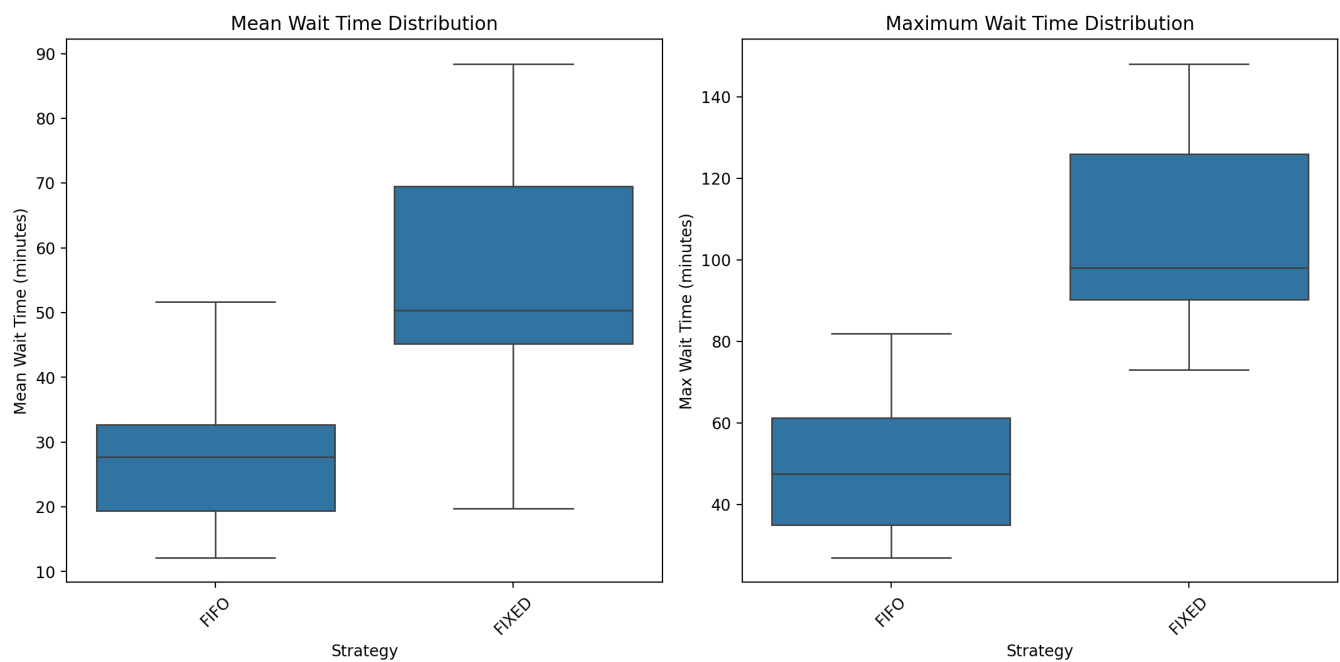


Figure 2. Wait-time distribution comparison across strategies.

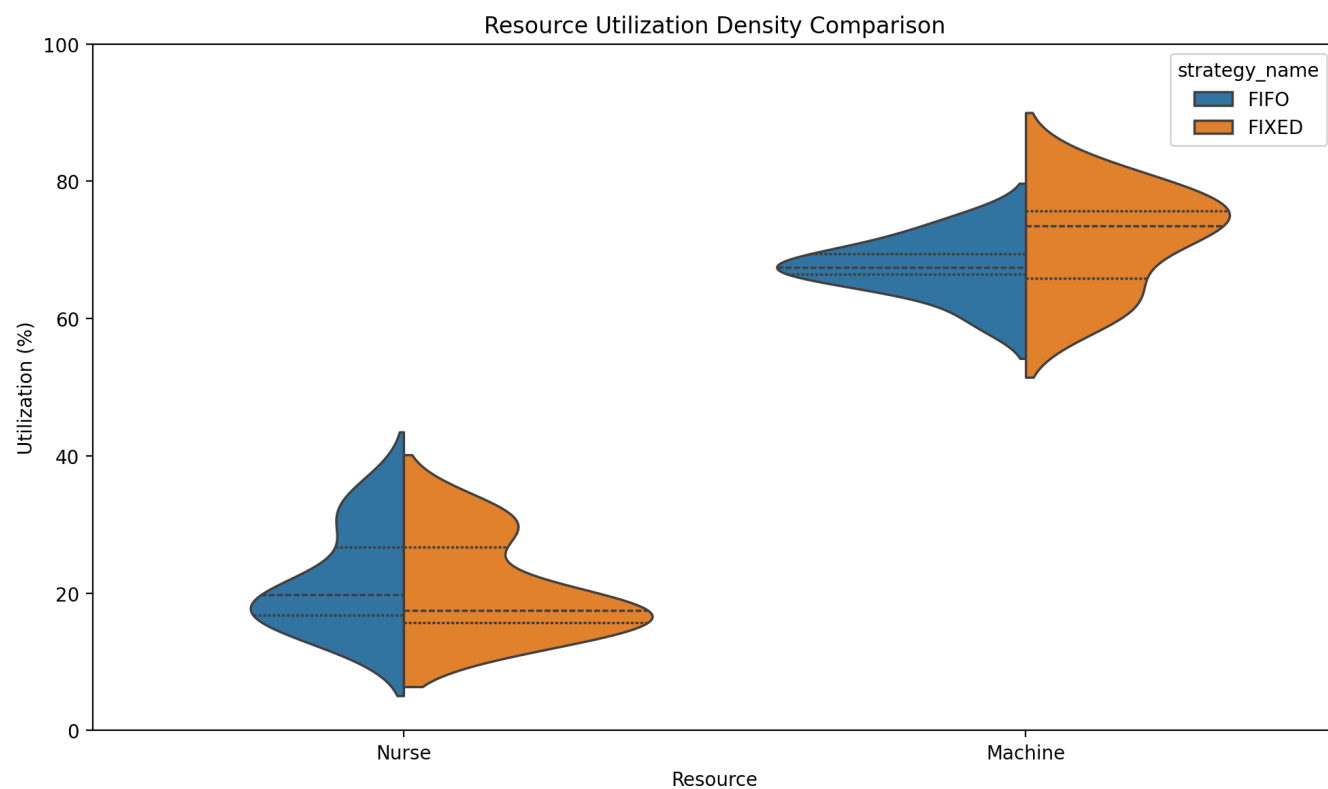


Figure 3. Utilisation violin plots across strategies.

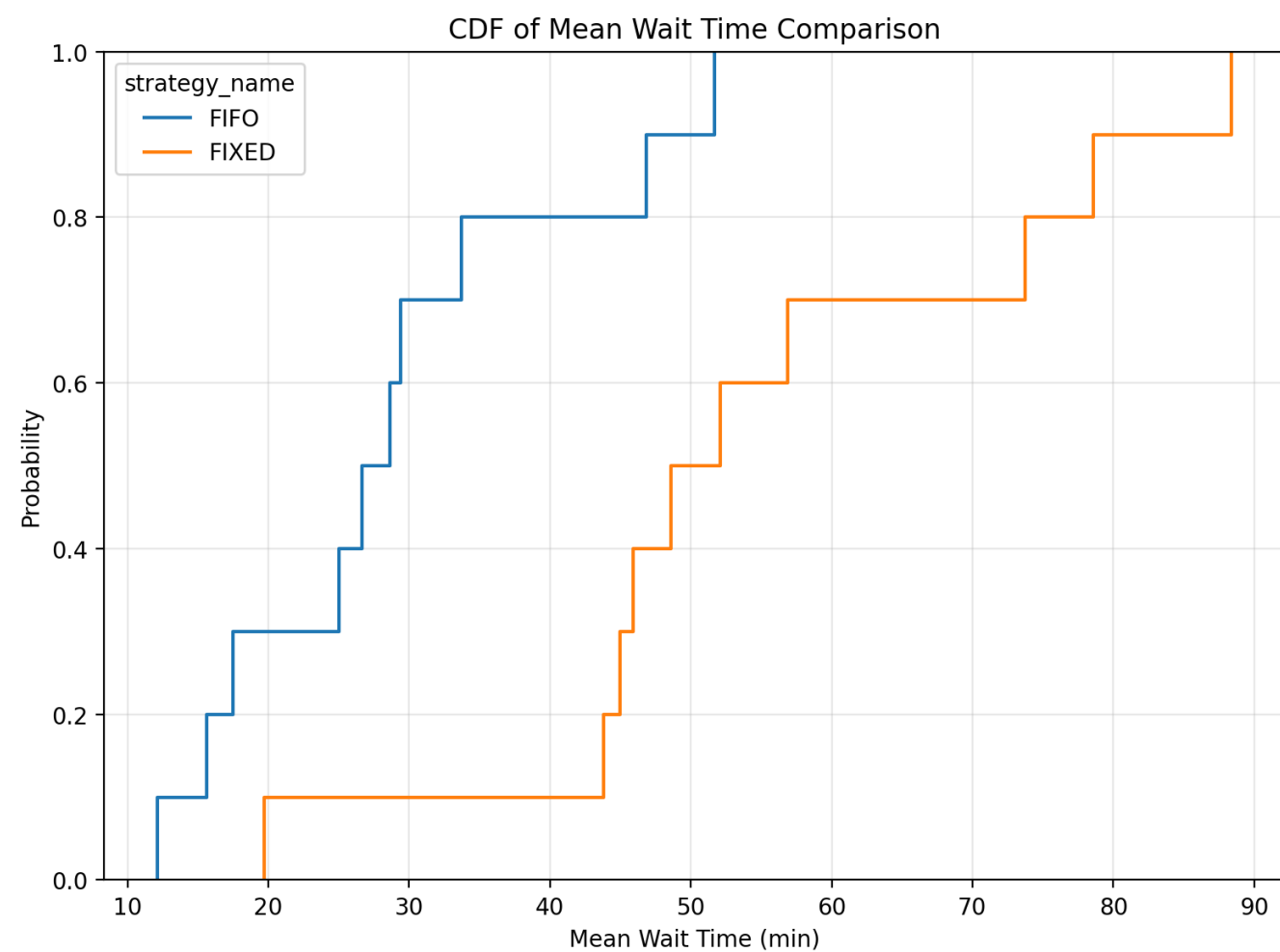


Figure 4. Empirical CDF of mean wait time -- all strategies.

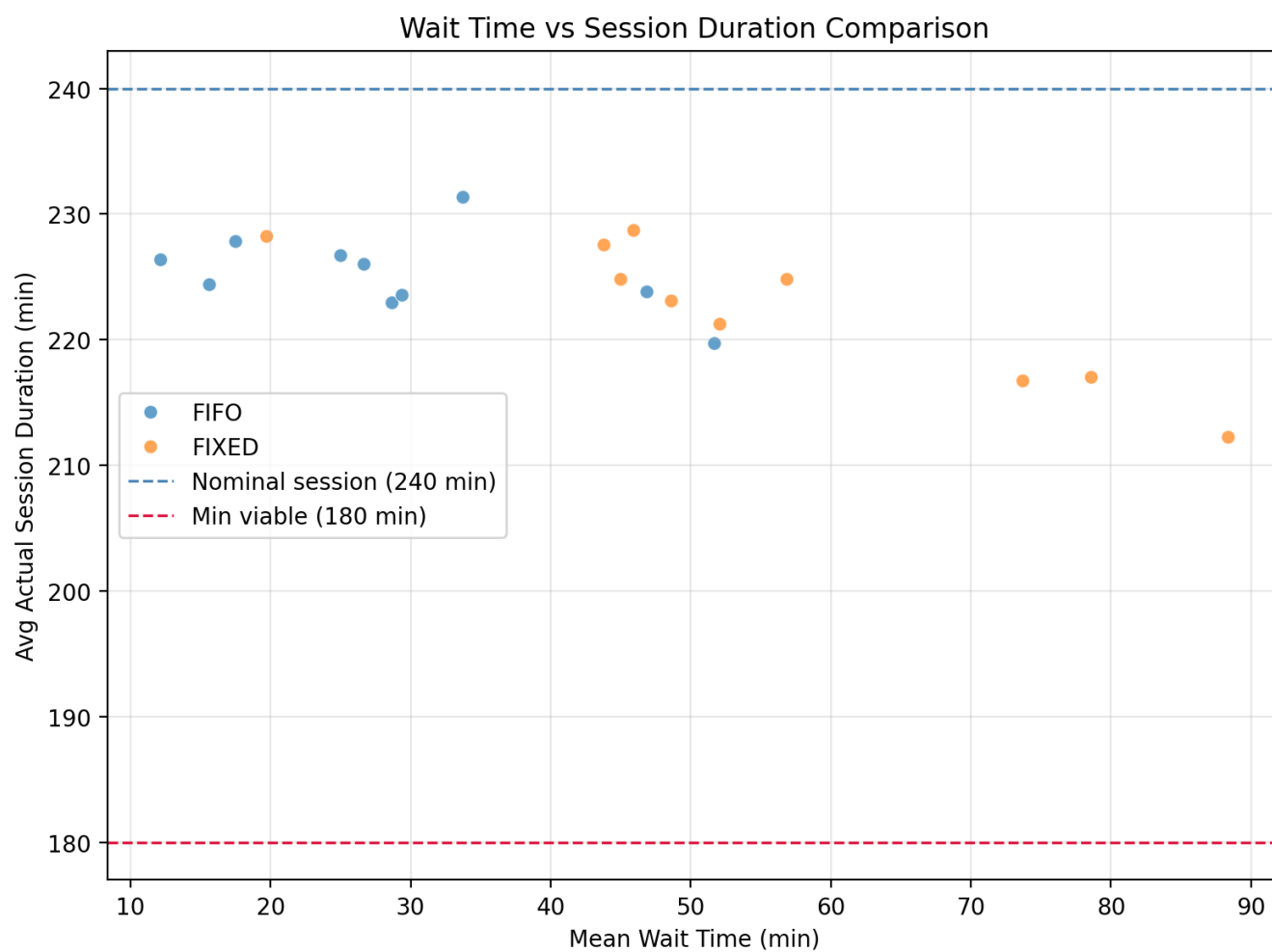


Figure 5. Wait time vs. avg session duration -- all strategies.

4. Edge-Case Stress Tests

Performance under extreme or auto-captured boundary conditions. Each card shows the full shift snapshot followed by per-strategy outcomes.

Scenario: Burst Arrival

Shift Snapshot

Active machines	20
Defective machines	0
Total machines	20
Nurses	2
Patients	20
Arrival window	0 - 0 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	42

Strategy Results

[PASS] FIFO | Patients: 20 | Failed: 0 | Truncated: 4

Wait (mean/max): 66.5 / 139.0 min | Avg session: 220 min | Nurse util: 41.4% | Machine util: 83.6%

[PASS] FIXED | Patients: 20 | Failed: 0 | Truncated: 4

Wait (mean/max): 66.5 / 139.0 min | Avg session: 220 min | Nurse util: 41.4% | Machine util: 83.6%

Scenario: Maintenance Disaster

Shift Snapshot

Active machines	15
Defective machines	12
Total machines	27
Nurses	4
Patients	20
Arrival window	1 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	42

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 17 | Truncated: 0

Wait (mean/max): 0.0 / 0.0 min | Avg session: 215 min | Nurse util: 3.1% | Machine util: 63.9%

[FAIL] FIXED | Patients: 20 | Failed: 17 | Truncated: 0

Wait (mean/max): 0.0 / 0.0 min | Avg session: 228 min | Nurse util: 2.6% | Machine util: 66.8%

Scenario: Marathon Shift

Shift Snapshot

Active machines	15
Defective machines	0
Total machines	15
Nurses	2
Patients	20
Arrival window	10 - 200 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	42

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 5 | Truncated: 5

Wait (mean/max): 0.0 / 0.0 min | Avg session: 230 min | Nurse util: 41.7% | Machine util: 69.4%

[FAIL] FIXED | Patients: 20 | Failed: 5 | Truncated: 5

Wait (mean/max): 0.0 / 0.0 min | Avg session: 230 min | Nurse util: 41.7% | Machine util: 69.4%

Scenario: Auto: seed-0 (FIFO)

Shift Snapshot

Active machines	13
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Defective machines	2
Total machines	15
Nurses	3
Patients	18
Arrival window	2 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	0

Strategy Results

[FAIL] FIFO | Patients: 18 | Failed: 5 | Truncated: 0

Wait (mean/max): 25.0 / 48.0 min | Avg session: 227 min | Nurse util: 19.6% | Machine util: 67.9%

Scenario: Auto: seed-0 (FIXED)

Shift Snapshot

Active machines	13
Defective machines	2
Total machines	15
Nurses	3
Patients	18
Arrival window	2 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	0

Strategy Results

[FAIL] FIXED | Patients: 18 | Failed: 5 | Truncated: 1

Wait (mean/max): 45.9 / 88.0 min | Avg session: 229 min | Nurse util: 18.1% | Machine util: 75.6%

Scenario: Auto: seed-1 (FIXED)

Shift Snapshot

Active machines	17
Defective machines	1
Total machines	18
Nurses	4
Patients	16
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	1

Strategy Results

[FAIL] FIXED | Patients: 16 | Failed: 1 | Truncated: 0

Wait (mean/max): 43.8 / 94.0 min | Avg session: 228 min | Nurse util: 15.2% | Machine util: 64.3%

Scenario: Auto: seed-2 (FIXED)**Shift Snapshot**

Active machines	15
Defective machines	1
Total machines	16
Nurses	2
Patients	15
Arrival window	1 - 59 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	2

Strategy Results

[FAIL] FIXED | Patients: 15 | Failed: 1 | Truncated: 7

Wait (mean/max): 73.7 / 131.0 min | Avg session: 217 min | Nurse util: 31.7% | Machine util: 74.7%

Scenario: Auto: seed-3 (FIXED)**Shift Snapshot**

Active machines	18
Defective machines	1
Total machines	19
Nurses	4
Patients	16
Arrival window	9 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	3

Strategy Results

[FAIL] FIXED | Patients: 16 | Failed: 1 | Truncated: 0

Wait (mean/max): 45.0 / 89.0 min | Avg session: 225 min | Nurse util: 15.5% | Machine util: 60.4%

Scenario: Auto: seed-4 (FIFO)**Shift Snapshot**

Active machines	15
Defective machines	3
Total machines	18
Nurses	3
Patients	16
Arrival window	1 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	4

Strategy Results**[FAIL] FIFO | Patients: 16 | Failed: 1 | Truncated: 0**

Wait (mean/max): 28.7 / 38.0 min | Avg session: 223 min | Nurse util: 19.8% | Machine util: 66.5%

Scenario: Auto: seed-4 (FIXED)**Shift Snapshot**

Active machines	15
Defective machines	3
Total machines	18
Nurses	3
Patients	16
Arrival window	1 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	4

Strategy Results**[FAIL] FIXED | Patients: 16 | Failed: 3 | Truncated: 0**

Wait (mean/max): 48.6 / 96.0 min | Avg session: 223 min | Nurse util: 16.2% | Machine util: 62.0%

Scenario: Auto: seed-5 (FIFO)**Shift Snapshot**

Active machines	15
Defective machines	1
Total machines	16
Nurses	3
Patients	19
Arrival window	0 - 56 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	5

Strategy Results**[FAIL] FIFO | Patients: 19 | Failed: 4 | Truncated: 0**

Wait (mean/max): 26.7 / 47.0 min | Avg session: 226 min | Nurse util: 19.8% | Machine util: 67.2%

Scenario: Auto: seed-5 (FIXED)**Shift Snapshot**

Active machines	15
Defective machines	1
Total machines	16
Nurses	3
Patients	19
Arrival window	0 - 56 min

Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	5

Strategy Results

[FAIL] FIXED | Patients: 19 | Failed: 4 | Truncated: 1

Wait (mean/max): 56.9 / 111.0 min | Avg session: 225 min | Nurse util: 19.9% | Machine util: 75.7%

Scenario: Auto: seed-6 (FIFO)

Shift Snapshot

Active machines	15
Defective machines	2
Total machines	17
Nurses	2
Patients	19
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	6

Strategy Results

[FAIL] FIFO | Patients: 19 | Failed: 4 | Truncated: 0

Wait (mean/max): 33.7 / 65.0 min | Avg session: 231 min | Nurse util: 31.5% | Machine util: 72.2%

Scenario: Auto: seed-6 (FIXED)

Shift Snapshot

Active machines	15
Defective machines	2
Total machines	17
Nurses	2
Patients	19
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	6

Strategy Results

[FAIL] FIXED | Patients: 19 | Failed: 4 | Truncated: 7

Wait (mean/max): 78.6 / 146.0 min | Avg session: 217 min | Nurse util: 29.9% | Machine util: 81.0%

Scenario: Auto: seed-7 (FIFO)

Shift Snapshot

Active machines	13
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Defective machines	2
Total machines	15
Nurses	2
Patients	17
Arrival window	2 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	7

Strategy Results**[FAIL] FIFO | Patients: 17 | Failed: 4 | Truncated: 0**

Wait (mean/max): 51.7 / 82.0 min | Avg session: 220 min | Nurse util: 29.0% | Machine util: 69.9%

Scenario: Auto: seed-7 (FIXED)**Shift Snapshot**

Active machines	13
Defective machines	2
Total machines	15
Nurses	2
Patients	17
Arrival window	2 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	7

Strategy Results**[FAIL] FIXED | Patients: 17 | Failed: 4 | Truncated: 6**

Wait (mean/max): 88.4 / 148.0 min | Avg session: 212 min | Nurse util: 29.0% | Machine util: 78.6%

Scenario: Auto: seed-8 (FIFO)**Shift Snapshot**

Active machines	13
Defective machines	3
Total machines	16
Nurses	3
Patients	16
Arrival window	1 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	8

Strategy Results**[FAIL] FIFO | Patients: 16 | Failed: 3 | Truncated: 0**

Wait (mean/max): 29.4 / 50.0 min | Avg session: 224 min | Nurse util: 17.1% | Machine util: 66.5%

Scenario: Auto: seed-8 (FIXED)

Shift Snapshot	
Active machines	13
Defective machines	3
Total machines	16
Nurses	3
Patients	16
Arrival window	1 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	8

Strategy Results	
[FAIL] FIXED Patients: 16 Failed: 3 Truncated: 0	
Wait (mean/max): 52.1 / 100.0 min Avg session: 221 min Nurse util: 16.9% Machine util: 72.2%	

Scenario: Auto: seed-9 (FIFO)

Shift Snapshot	
Active machines	14
Defective machines	3
Total machines	17
Nurses	4
Patients	18
Arrival window	5 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	9

Strategy Results	
[FAIL] FIFO Patients: 18 Failed: 4 Truncated: 0	
Wait (mean/max): 12.1 / 34.0 min Avg session: 226 min Nurse util: 14.1% Machine util: 67.6%	

Scenario: Auto: seed-9 (FIXED)

Shift Snapshot	
Active machines	14
Defective machines	3
Total machines	17
Nurses	4
Patients	18
Arrival window	5 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	9

Strategy Results**[FAIL] FIXED | Patients: 18 | Failed: 4 | Truncated: 0**

Wait (mean/max): 19.7 / 73.0 min | Avg session: 228 min | Nurse util: 14.9% | Machine util: 70.7%